

# **Dramatic Shifts Over Ocean for Biogeochemistry Components in CMIP6**

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November 5, 2025

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**1 Introduction**

**2 Data & Methodology**

### 3 Results

Below, we list the biogeochemistry components of the ocean-based climate systems that undergo dramatic shifts. These shifts can be classified as either gradual or abrupt transitions, depending on their evolution. Regardless of the type, we refer to them in this text as strong non-linear surprises (SNSs).

We apply the method developed by [Angevaere and Drijfhout \(2025\)](#) to identify SNSs in the ocean biogeochemistry components of CMIP6 models that provide data on these variables. The data include ocean surface chlorophyll concentration (chlos), ocean surface phytoplankton carbon concentration (phycos), ocean surface zooplankton carbon concentration (zoocos), ocean surface dissolved inorganic nitrate concentration (no3os), vertically integrated net primary production rate (intpp), and vertically integrated nitrogen fixation rate (intpn2). Because several of these metrics (the first five) show strong seasonality (as noticed by [Kelly et al. \(2025\)](#) as well) in the ocean, we separately apply criteria to the annual maximum of these metrics, similar to the approach of [Angevaere and Drijfhout \(2025\)](#) for the detection of shifts in the ocean mixed-layer depth. In the following, we report on the detected SNSs separately for northern North Atlantic Ocean (sec. 3.1), Arctic Ocean (3.2), and Indo-Pacific Ocean (sec 3.3).

#### 3.1 Northern North Atlantic

Over the northern region of the North Atlantic Ocean, as shown in Fig. 1, four models project a gradual decrease in the annual maximum of no3os, intpp, chlos, and zoocos. In addition, one model shows an abrupt decrease in both chlos and phycos on the western side of Greenland, in the Labrador Sea. We attribute these dramatic shifts in this region to the shoaling of the mixed-layer depth and, in turn, to the slowdown of the AMOC. This process reduces the supply of nutrients to the ocean surface, as diminished vertical mixing decreases the transport of nutrients from deeper layers. The resulting nutrient limitation leads to a decline in net primary production rates, and consequently in chlos, phycos, and zoocos (e.g., [Kelly et al., 2025](#)).

#### Phycos-max, chlos-max, zoocos-max, intpp-max, DIN-max decrease in North Atlantic

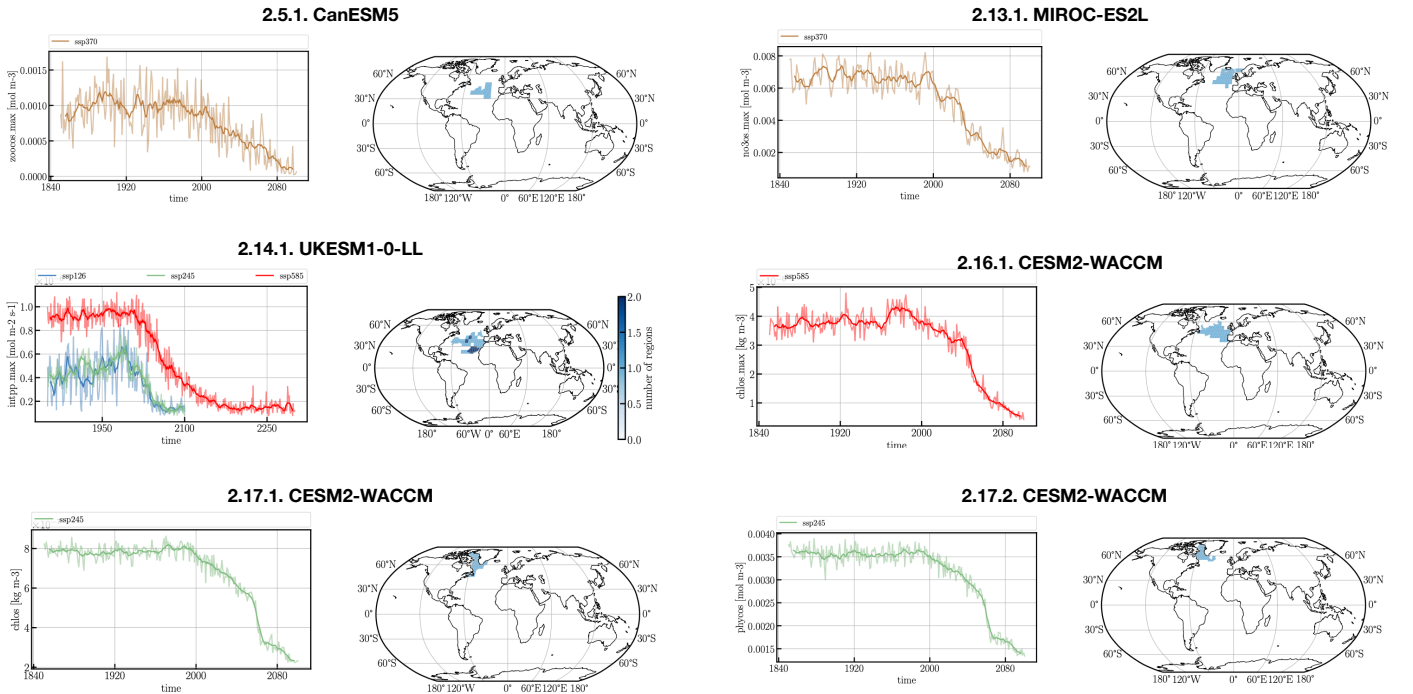


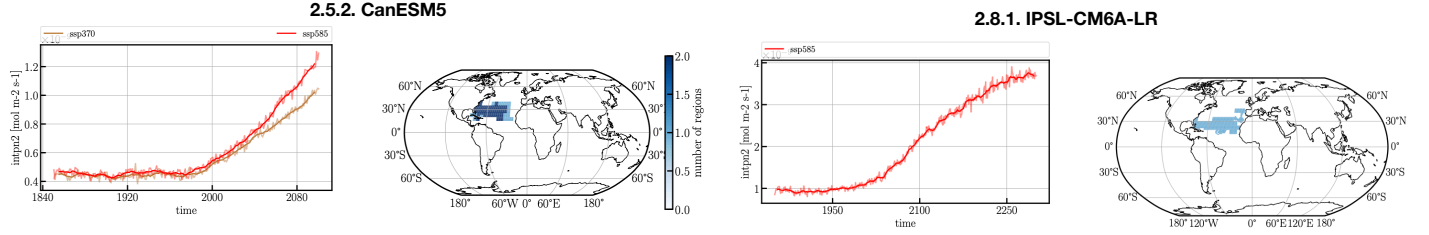
Figure 1: Time series of the SNSs for the Northern North Atlantic Ocean for the gradual decrease in annual maximum zoocos, no3os, intpp, and chlos, and the abrupt decrease in the yearly mean chlos and phycos.

In addition, two models in Fig. 2 project a gradual increase in intpn2 over the subtropical North Atlantic. If intpn2 is primarily temperature-dependent ([Deutsch et al., 2024](#)), this increase may reflect the influence of rising temperatures up to a favorable amount. Nevertheless, the underlying mechanism driving this shift in intpn2 in these two models remains

unclear.

Moreover, the yearly maximum intpp is projected to increase over a relatively small region east of Barbados in the IPSL-CM6A-LR model (Fig. 2). If primary production depends on vertical mixing and deep-water formation, this increase in intpp could be a consequence of enhanced mixing in that area. However, further investigation is needed to better understand the possible mechanisms behind this gradual increase in intpp.

## Intpn2 increase in North Atlantic



## Intpp-max increase in North Atlantic

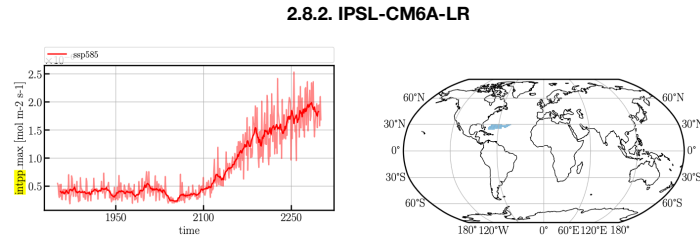


Figure 2: Time series of the SNSs for the Northern North Atlantic Ocean for the gradual increase in yearly mean intpn2, and annual maximum of intpp.

## 3.2 Arctic

Three models (Fig. 3) project a gradual increase in the yearly maximum of chlos, phycos, zoocos, and intpp over the Arctic Ocean. We attribute these shifts to the decrease in sea-ice extent, which leads to a greater amount of absorbed sunlight and, in turn, enhances primary production that drives increases in chlos, phycos, and zoocos. In contrast to these models, the UKESM1 model (Fig. 4) projects a gradual decrease in chlos and phycos over the Arctic region. This contrast may be explained by the fact that, over the Arctic, the UKESM1 model shows a strong decrease in no3os—from around 15 to 5 mmol/m<sup>3</sup> (Fig. 5)—which drives the decline in intpp, chlos, and phycos. However, in models such as MIROC-ES2L, despite a decrease of about 4 mmol/m<sup>3</sup> in no3os (Fig. 5), this reduction does not appear strong enough to offset the impacts of increased sunlight on chlos, phycos, and intpp growth.

# Increase in intpp-max, chlos-max, phycos-max, zoocos-max over Arctic

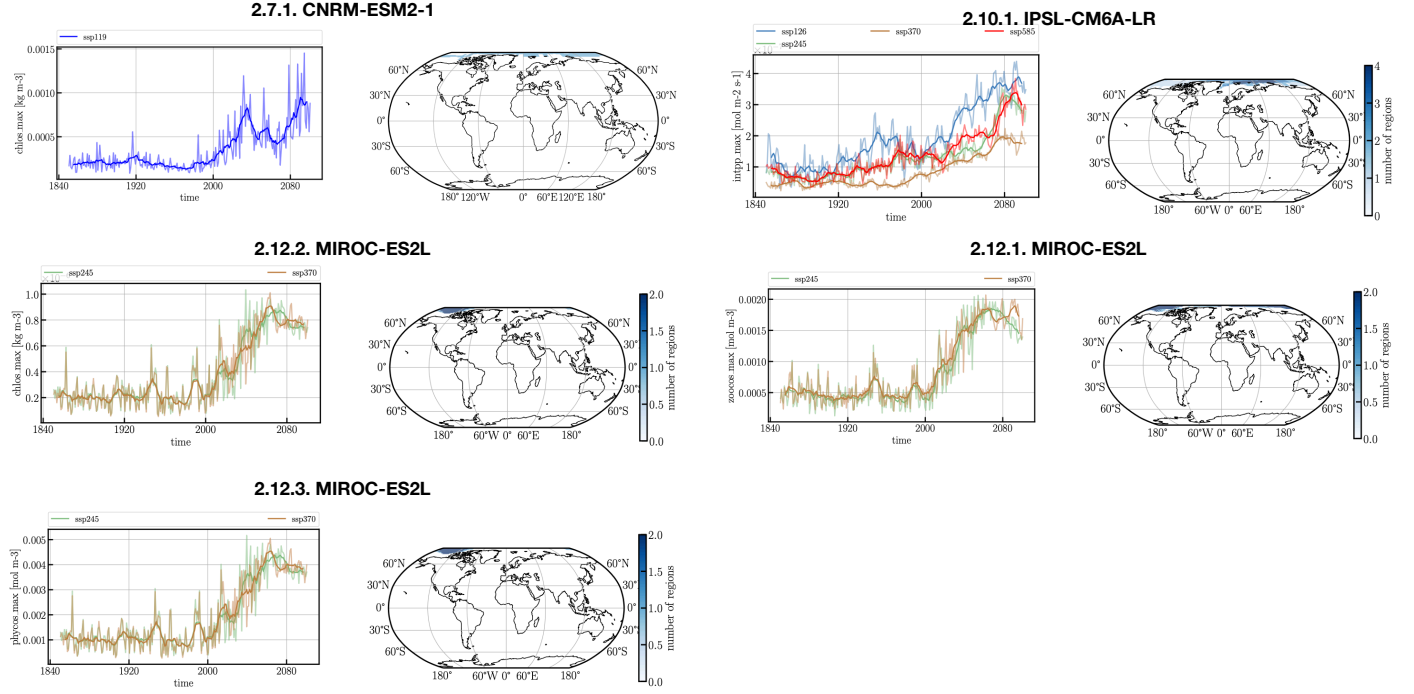


Figure 3: Time series of the SNSs for the Arctic Ocean for the gradual increase in annual maximum zoocos, intpp, and chlos, and phycos.

## Chlos/phycos-max decrease over Arctic

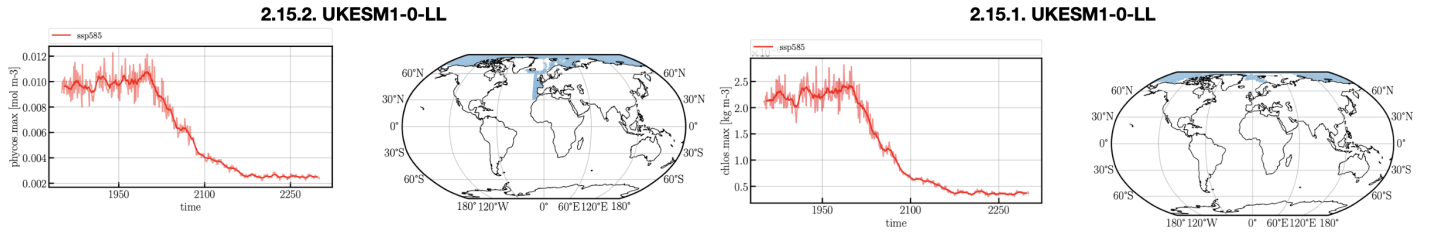


Figure 4: Time series of the SNSs for the Arctic Ocean for the gradual decrease in annual maximum chlos and phycos.

## DIN-max decrease over Arctic

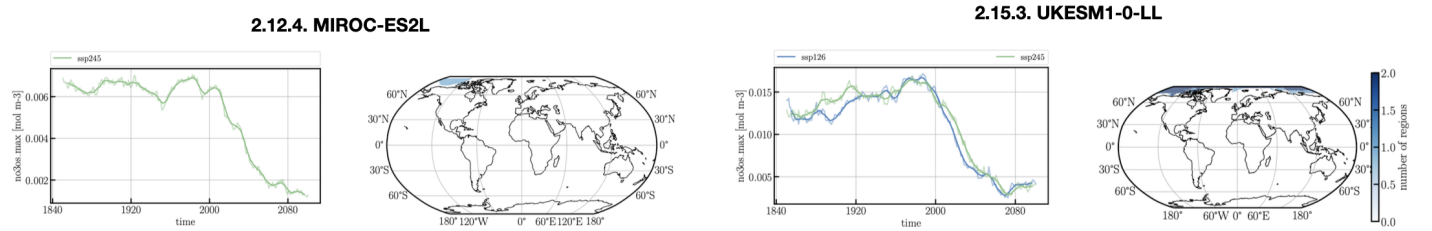


Figure 5: Time series of the SNSs for the Arctic Ocean for the gradual decrease in annual maximum no3os.

### 3.3 Indo-Pacific

Two models project a gradual decrease in no3os on the western side of South America (Fig. 6). This could be due to weakened westerlies resulting from these models transitioning to a mean El-Niño-like state (e.g., [Erickson and Patricola, 2023](#)). Such a situation corresponds to reduced westerly winds, which in turn drive less upwelling of water from deeper ocean layers, ultimately reducing the amount of nutrients such as no3os at the ocean surface.

## DIN-max decrease over central Pacific Ocean

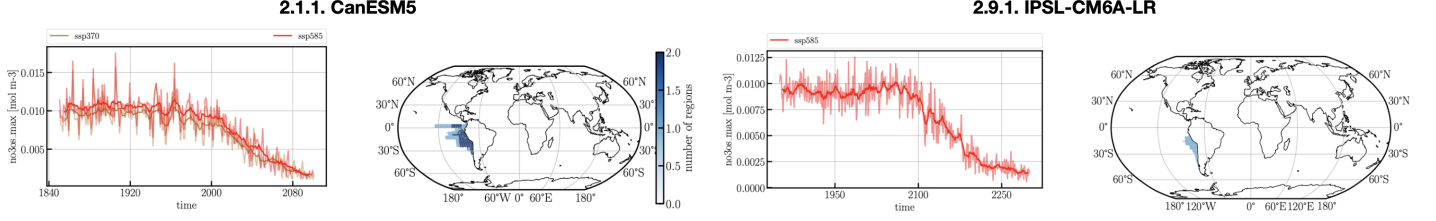


Figure 6: Time series of the SNSs for the central Pacific Ocean for the gradual decrease in annual maximum no3os.

Furthermore, the CanESM5 model projects a gradual decrease in the yearly maximum zoocos over three regions (Fig. 7): the (sub)tropics, and around 30°N and 30°S. We currently do not know of a specific mechanism that could drive such a change. In addition, there is an abrupt decrease in phycos projected by the IPSL-CM6A-LR model over the tropical region of the central Pacific (Fig. 7), for which we do not yet have an explanation. Moreover, the CanESM5 model projects a gradual increase in intpn2 over a region of the Pacific Ocean at 30°S latitude (Fig. 7), for which we likewise do not have a clear physical explanation.

## Phycos and zoocos-max decrease over Pacific Ocean

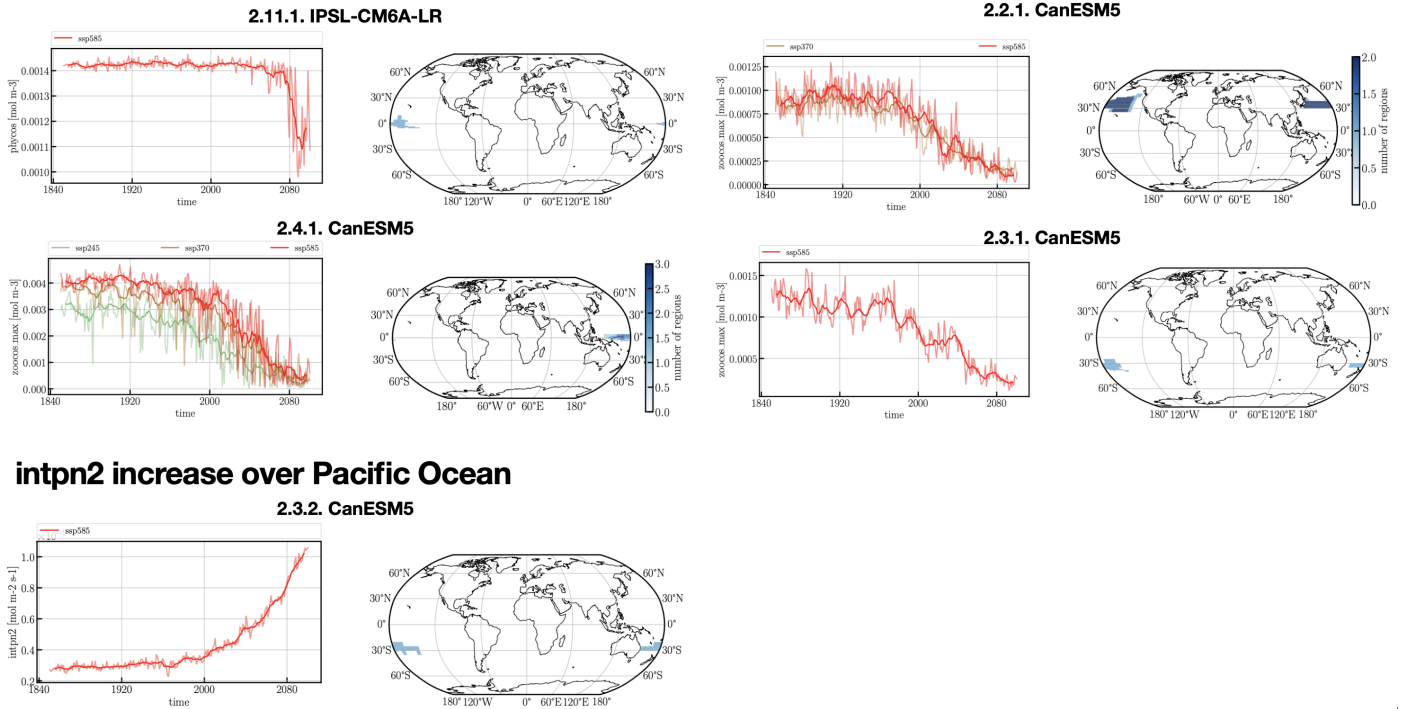


Figure 7: Time series of the SNSs for the Pacific Ocean for the gradual decrease in annual maximum zoocos, an abrupt decrease in yearly mean phycos, and a gradual increase in yearly mean intpn2.

The CMCC-ESM2 model projects a gradual increase in the yearly maximum no3os south of India in the Indian Ocean (Fig. 8). This might be related to decreased ocean mixing over the region, although we currently do not know the possible drivers of such a transition.

### 2.6.1. CMCC-ESM2

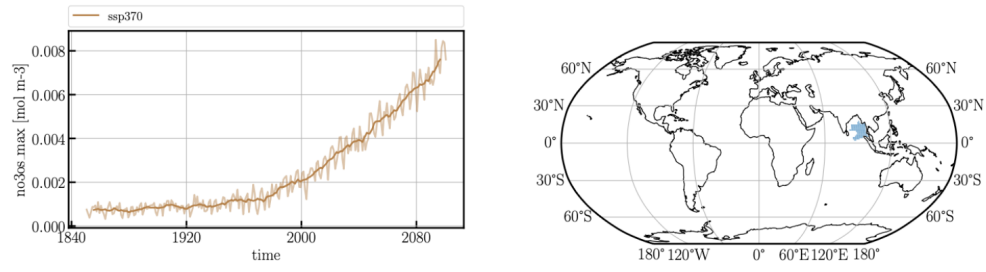


Figure 8: Time series of the SNSs for the Indian Ocean for the gradual increase in annual maximum no3os.

## 4 Summary & Conclusion

## References

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